

PerChannelMinMaxObserver#

[https://pytorch.org/docs/stable/generated/torch.ao.quantization.observer.Per](https://pytorch.org/docs/stable/generated/torch.ao.quantization.observer.PerChannelMinMaxObserver.html)

Description:

Observer module for computing the quantization parameters based on the running per channel min and max values. This observer uses the tensor min/max statistics to compute the per channel quantization parameters. The module records the running minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters.

- ch_axis – Channel axis
- dtype – dtype argument to the quantize node needed to implement the reference model spec.
- qscheme – Quantization scheme to be used

Function Signature

```
class
torch.ao.quantization.observer.PerChannelMinMaxObserver(ch_a
xis=0, dtype=torch.quint8, qscheme=torch.per_channel_affine,
reduce_range=False, quant_min=None, quant_max=None,
factory_kwargs=None, eps=1.1920928955078125e-07,
is_dynamic=False, **kwargs)[source]#

Parameters

reset_min_max_vals()[source]#
```

Notes & Warnings

NOTE

Note If the running minimum equals to the running maximum, the scales and zero_points are set to 1.0 and 0.

Detailed Documentation

Documentation

Observer module for computing the quantization parameters based on the running per channel min and max values.

This observer uses the tensor min/max statistics to compute the per channel

quantization parameters. The module records the running minimum and maximum of incoming tensors, and uses this statistic to compute the quantization parameters.

`ch_axis` – Channel axis

`dtype` – dtype argument to the quantize node needed to implement the reference model spec.

`qscheme` – Quantization scheme to be used

`reduce_range` – Reduces the range of the quantized data type by 1 bit

`quant_min` – Minimum quantization value. If unspecified, it will follow the 8-bit setup.

`quant_max` – Maximum quantization value. If unspecified, it will follow the 8-bit setup.

`eps (Tensor)` – Epsilon value for float32, Defaults to `torch.finfo(torch.float32).eps`.

The quantization parameters are computed the same way as in `MinMaxObserver`, with the difference that the running min/max values are stored per channel. Scales and zero points are thus computed per channel as well.

If the running minimum equals to the running maximum, the scales and zero_points are set to 1.0 and 0.

Resets the min/max values.